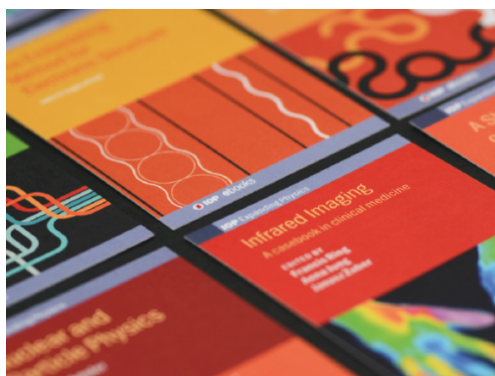


PAPER

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Experimentally measured phonon spectrum of microtubules

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Abstract

Microtubules are self-assembled protein cylinders with important mechanical roles inside living cells. Here we present an experimental method to measure the transversal collective dynamic modes of taxol stabilized microtubules *in vitro*. We treat the microtubule as a lattice of particles interacting with harmonic potentials and we use Fourier decomposition of the shape of the microtubule under excitation from thermal energy. We report the phonon spectrum due to these modes in MHz frequencies and the sound velocity is in the order of mm s^{-1} .

Keywords: topological phonons, biomechanics, microtubules, persistence length, phonon spectrum

 Supplementary material for this article is available [online](#)

(Some figures may appear in colour only in the online journal)

Microtubules are self-assembled protein lattices with unique mechanical properties. They have a cylindrical shape with a 25 nm outer diameter and can be tens to hundreds of microns long. Their mechanical properties are crucial in many cellular activities. Their functions are performed not by a static structure but by a highly dynamic one that constantly assembles and disassembles while maintaining the shape of the cell through a process known as dynamical instability (DI) [1]. The mechanism behind DI, still not understood, has been the focus of intense research over the past few decades.

It was previously proposed that microtubules are topological phononic crystals and that the mechanism behind the DI of microtubules could be based on topological resonant states at the edge, also known as edge modes [2]. These edge modes would cause large amplitude vibrations at the ends of the microtubules. This idea augments the theory of the ‘GTP cap’ model for DI. The ‘GTP cap’ model proposes that a few rows of tubulin dimers with unhydrolyzed guanosine triphosphate (GTP) are capable of stabilizing the whole microtubule, and that there is an energy barrier that must be overcome in order for the growing microtubule to transition into a shortening microtubule. The edge mode of the microtubule could provide this energy by storing vibrational energy at the growing ends of the microtubule. The GTP hydrolysis at the ends adds

energy to the system, and the edge mode would be amplified until finally the vibrations are enough to overcome the energy barrier between the growing and the shortening phases.

Measuring the phonon spectrum is a standard measurement in materials science to understand the dynamics of the underlying mechanical structure of a material. However, since microtubules have a short life span (less than a few minutes), exist in an aqueous environment, and are very sensitive to changes in the chemical environment, this makes it difficult to apply standard methods to measure the phonon spectrum.

In this paper we present a method to measure the phonon spectrum in order to generate a microscopic picture of the collective transversal vibrational modes of the microtubule, i.e. those which predominantly involve the motion of the tube perpendicular to the length of the microtubule. The method is non-invasive and the data can be collected while the microtubules are *in vitro*, which mimics their natural environment. The focus of the method is to shift from the static to the dynamic characteristics of these unique self-assembled structures. To exemplify the method, we use the thermal energy of water to excite the natural frequencies of microtubules and record their motion in fluorescence using a high-speed camera. The method could be further enhanced to provide a window into the local mechanical characteristics of microtubules, such

as the protein–protein force fields, at scales fine enough to detect the presence and the effect of various proteins, such as motor proteins, and chemical compounds, such as the chemotherapy drug Taxol. Furthermore, the latest developments in theoretical modeling and sensing of metamaterials has shown promise for the ability to design and engineer materials that exhibit unique surface and edge states. Studying the exotic properties of microtubules and nano-scale metamaterials is a promising field of study that has applications in high-speed computing, energy harvesting, sensing, and model based design of new materials [12–16].

Pure double-cycled tubulin and tubulin labeled with Alexa488 and Alexa594 were stored in -80°C in PEM buffer (80 mm piperazine-N,N'-bis(2-ethanesulfonic acid) (PIPES), 1 mm ethylene glycol tetraacetic acid (EGTA), 1 mm MgCl_2 , pH 6.9).

To grow microtubules the tubulin was thawed on ice and grown in a GPEM buffer (1 mm guanine triphosphate (GTP), 80 mm PIPES, 1 mm EGTA, 1 mM MgCl_2 , pH 6.9) with 10% dimethyl sulfoxide. The solution was incubated on ice for 5 min and then incubated at 37°C for one hour. The tubulin was grown at 2 mg ml^{-1} with 10%–50% labeled tubulin. After incubation, warm Taxol was added to a final concentration of $10\text{ }\mu\text{M}$. The stabilized microtubules were diluted 100 times in an anti-bleaching solution (250 nm glucose oxidase, 64 nm catalase, 40 mm D-glucose, 1% beta-mercaptoethanol) before imaging [3]. Tubulin was purchased from PurSolutions. All chemicals were purchased from Fisher Scientific. Taxol was purchased from Cytoskeleton, Inc.

Samples for imaging were prepared by coating the glass slide and coverslip with bovine serum albumin, dropping $1.5\text{--}2\text{ }\mu\text{l}$ of the diluted microtubule solution onto the slide, and sealing the coverslip to the slide with 1 min epoxy.

A Prime 95BTM Scientific CMOS Camera was used to acquire videos of the microtubules at 10–100 frames per second.

The plug-in JFilament in FIJI was used to find the initial coordinates of the microtubule [4]. The ends of the microtubule can easily move out of focus, and this results in missing data points in the tracking. Therefore, only microtubules that stayed in focus for 1000 frames were used for analysis to ensure that the length of the tracked microtubule did not change significantly over time as shown in figure 1.

The coordinates from JFilament were then fit to a spline in MATLAB, and segmented into 26, 50, or 100 equal length blocks as shown in figure 2. Each block is represented by the position of its center of mass as shown in figure 3. To ensure that a single microtubule was measured and not a bundle, their persistence length was measured using standard methods [5]. The coordinates from the microtubule tracking as described above were then used to determine the phonon spectrum.

Generic mechanical systems are made of rigid blocks interacting via force fields. In microtubules, these blocks are the tubulin dimers themselves, but, as we shall see later, at different levels of characterization the blocks can take other meanings. The force fields are the interaction

between the proteins that keeps the microtubule together. Thus we model the microtubule as a discrete system of N blocks interacting with harmonic potentials. This system has a large number of degrees of freedom leading to a large number of natural collective modes. The actuation of the system is done through a force generated by the thermal energy. This is a time-varying force that is not tuned at all and excites all the collective modes at once with different weights.

The displacements are the response of the microtubule to the thermal energy forces buffeting it around in the solution. The interactions between the blocks of the microtubule are represented by a harmonic potential, where K_n is the stiffness matrix. The thermal energy is derived from the equipartition theorem, $\frac{1}{2}k_B T$, where k_B is the Boltzmann constant and T is temperature. The thermal energy and the microtubule are in equilibrium, therefore:

$$V_{\text{thermal}} = V_{\text{blocks}} = \frac{1}{2}k_B T = \frac{1}{2} \sum_{n=1}^N K_n u_t(n)^2. \quad (1)$$

In order to glean the material properties from this equilibrium assumption, first, the displacement of each coordinate was calculated in reference to an average position to get N positions for each frame, where N is equal to the number of coordinates:

$$u_t(n), n = 1, \dots, N \quad t \in [0, T]. \quad (2)$$

At a fixed time, t , the shape of the microtubule can be modeled by a fourier approximation of these displacements:

$$u_t(n) = u_t(q_0) + \sum_{q=1}^N e^{i2\pi qn/N} u_t(q). \quad (3)$$

A Fourier transform is used to solve for the unknown fourier coefficients and to go from n to q (mode number). For any fixed time, t , we can encode the information in:

$$\tilde{u}_t(q) = \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{i2\pi qn/N} u_t(n) \quad q = 1, \dots, N. \quad (4)$$

Using $\tilde{u}_q$'s we can write the energy in a completely equivalent way:

$$U = \frac{1}{2} \sum_{q=1}^N F(q) (\tilde{u}(q) - \tilde{u}_0(q))^2, \quad (5)$$

where

$$F(q) = \sum_{n=1}^N e^{i2\pi qn/N} K_n. \quad (6)$$

The function $F(q)$ can be found from the variance of the Fourier coefficients, within which is encoded information about the stiffness matrix and therefore the resonant frequencies of the microtubule can be estimated.

$$\langle (\tilde{u}_q(t + \Delta t) - \tilde{u}_q(t))^2 \rangle_{t, \Delta t > \tau} = 2k_B T / F(q). \quad (7)$$

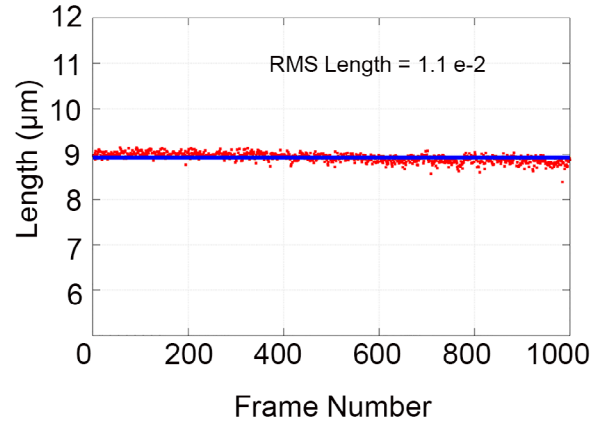


Figure 1. The length of the tracked microtubule was found by summing the length of each block. The length was measured for each frame.

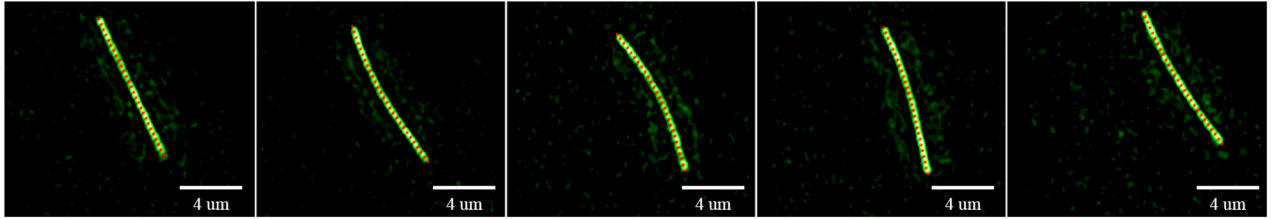


Figure 2. Tracking of microtubule (green) segmented into 26 blocks. Each point (red) represents the position of the center of mass of that block.

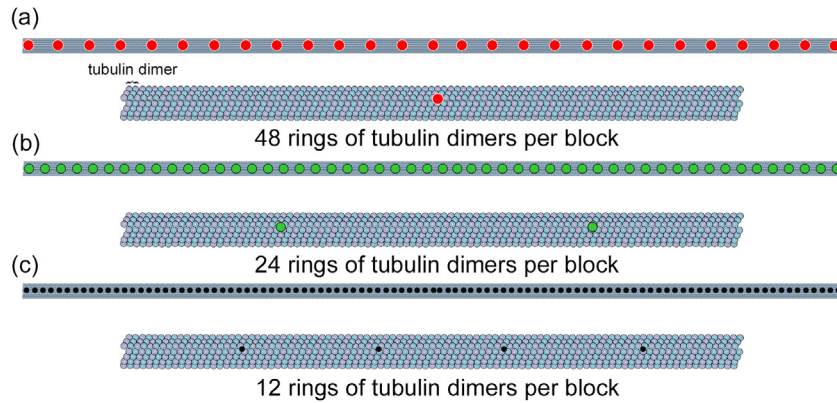


Figure 3. The points along the microtubule represent the center of mass of each block. (a) A microtubule segmented into 26 blocks with a point representing the position of center of mass. (b) A microtubule with 50 blocks. (c) A microtubule with 100 blocks.

Thus the angular frequency can be extracted using the formula below:

$$\omega_q = \sqrt{F(q)/M}, \quad (8)$$

where M is the mass of individual tubulin dimers. Finally, the phonon spectrum is plotted as a function of frequency, $f = \frac{\omega_q}{2\pi}$, versus the wave vector, $k = \frac{q\pi}{N}$.

In the long wave limit $\lim_{k \rightarrow 0}$ the slope of the linear region of the phonon spectrum represents speed of sound and is an indicator of the stiffness of the microtubule.

$$v = \lim_{k \rightarrow 0} \frac{\delta\omega}{\delta k}. \quad (9)$$

Ideally N would be equal to the number of dimers in a microtubule, but this is not possible due to the resolution of the microscope. As described in figure 3 with a coarse

discretization of $N = 26$, where one block represents 48 rings of tubulin dimers, we obtain 26 modes spread across the whole phonon spectrum. By defining the discretization and increasing N , such that $N = 50$ and 100, the results will represent the reality more accurately. When N increases, the spacing between the natural frequencies decrease and more modes will fill in the phonon spectrum.

Next, to ensure that we measure single microtubules, we use a set of standard techniques to determine the flexural rigidity and persistence length and compare it to the ones in the literature. Here it is assumed that the microtubule behaves as a rigid rod. The flexural rigidity and persistence length were measured by approximating the microtubule's curvature, $\theta(s)$, as a Fourier cosine series where $\theta(s)$ is the angle of each block to the horizontal, s is the block length, and L is the total length of the microtubule [5]:

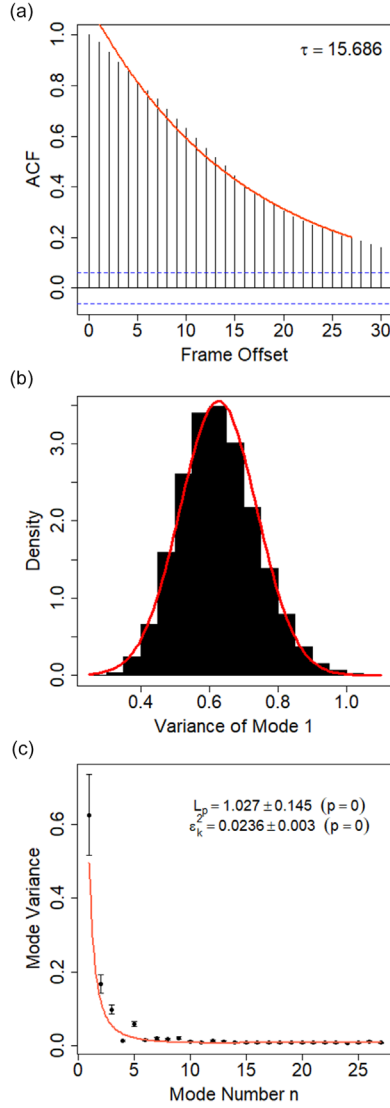


Figure 4. (a) Autocorrelation of the variances of the first mode where τ is the decay constant of the exponential fit. The independent number of frames is equal to $N_{ind} = \frac{N_{total}}{\tau}$. (b) A histogram of the variances of the first mode show a normal distribution showing that the bootstrapping method removes the bias introduced by oversampling. (c) The variance for each mode is plotted, and fit to a multiple linear regression model using equation (12).

$$\frac{k_B T}{EI} \left(\frac{L}{q\pi} \right)^2 = \langle (a_q(t + \Delta t) - a_q(t))^2 \rangle_{t, \Delta t > \tau} = \text{var}(a_q). \quad (10)$$

The persistence length is defined as $L_p = \frac{EI}{k_B T}$, equation (10), therefore:

$$\text{var}(a_q) = \frac{1}{L_p} \left(\frac{L}{q\pi} \right)^2. \quad (11)$$

Equation (11) shows that one can obtain the persistence length of a polymer by measuring the variance over time of the Fourier coefficients, as long as the modes are uncorrelated.

Accounting for the noise term due to the resolution of the microscope and camera, equation (11) becomes:

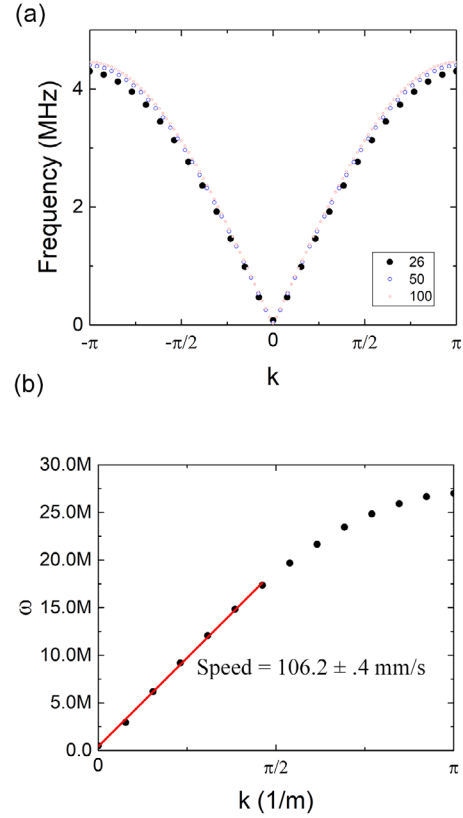


Figure 5. (a) The frequencies of the microtubule are plotted against the wave vector, $k = \frac{q\pi}{N}$, where N is the total number of blocks. This is the phonon spectrum of a microtubule segmented into 26, 50, and 100 blocks. (b) The slope of the dispersion, in the long wave limit, represents the speed of sound.

$$\text{var}(a_q) = \left(\frac{L}{q\pi} \right)^2 \frac{1}{L_p} + \frac{4}{L} \langle \epsilon_k^2 \rangle \left[1 + (N-1) \sin^2 \left(\frac{q\pi}{2N} \right) \right] \quad (12)$$

where ϵ_k is a fitting parameter that accounts for the experimental error [5].

The first few modes are prone to oversampling because of the frame rate of the video. To remove bias in the variance measurement due to oversampling, an autocorrelation function was used on the variance of the first mode to determine a correlation time, τ [6]. The correlation time was then used to find the number of time independent frames in the data set, $N_{ind} = \frac{\text{total number of frames}}{\tau}$. N_{ind} frames were then selected randomly and the variance was solved using equation (10). This was repeated 5000 times with replacement using a bootstrapping method coded in *R* to get a standard deviation for the variance of each Fourier coefficient. The variance of the Fourier coefficients was plotted against the mode number. Finally, multiple linear regression fitting was used to solve equation (12) for L_p and ϵ_k [6].

The average length of the microtubule over all frames was used in all calculations. There are small changes in length as the fluorescence intensity changes slightly for each frame because the ends and the microtubule fluorescence bleaches over time. Only microtubules that had a standard RMS deviation in length of less than 1% were used for analysis.

Table 1. This is a table of the persistence lengths and flexural rigidity of microtubules measured using the standard method described by equation (12). The analysis for the other microtubules are shown in detail in the supplemental material (stacks.iop.org/JPhysD/53/025401/mmedia).

Persistence length and flexural rigidity of microtubules				
	Frames per second	Length (μm)	Persistence length (mm)	Flexural rigidity (10^{23} N m^2)
MT 1	10 FPS	8.9	$1.8 \pm .5$	0.06 ± 0.02
MT 2	10 FPS	18.1	$1.0 \pm .1$	0.04 ± 0.01
MT 3	100 FPS	8.25	$.3 \pm .1$	0.01 ± 0.002
MT 4	100 FPS	17.8	$1.1 \pm .2$	0.04 ± 0.01

The persistence length of four microtubules was measured using the bootstrapping method as described in the methods and is shown in table 1. The persistence lengths were comparable to those found in the literature for free end microtubules [6]. The data indicates that we measured individual microtubules and that the tracking method is accurate.

For exemplification we present in figure 4 the data for a single $8.9 \mu\text{m}$ long microtubule. Figure 4(a) shows the autocorrelation on the variance of the first mode coefficient in the Fourier series up to 30 frames. The autocorrelation was fit to an exponential that gave a characteristic time constant used to determine the number of independent frames in a single data set of 1000 frames. The distribution in figure 4(b) of the variance of the first mode coefficient is Gaussian, indicating that the bootstrapping method removes bias in the variance measurement due to oversampling. Figure 4(c) shows the multiple linear regression fitting of equation (12) to the variance. The multiple linear regression model accounts for the large standard deviation of the first few modes. The first mode shows the largest standard deviation, which is expected as the first mode movement will always have the largest amplitude.

The phonon spectrum presented here is for the $8.9 \mu\text{m}$ long microtubule with a persistence length of 1.8 mm. The phonon spectrum was measured using the method described above. Figure 5(a) shows the phonon spectrum for the same microtubule segmented into 26, 50, and 100 blocks. The phonon spectrum is plotted as frequency versus $k = \frac{q\pi}{N}$ where N is the total number of segments along the microtubule. The plots show that increasing the segmentation fills in more resonant states for the phonon spectrum. Figure 5(b) shows the fitting of the spectrum in the long wave limit, from which the speed of sound is extracted.

The dynamicity of the microtubule has been observed and studied for decades, but the mystery of how local properties result in the final function of the microtubule is still a topic of intense research.

Currently, the understanding of DI as it functions normally is that it is a stochastic process due the stochastic nature of GTP hydrolysis. The GTP hydrolysis provides energy to the system to overcome the energy barrier between the growing and shortening phase of the MT, and built in strain energy in the microtubule allows fast shortening of the microtubules. However, this explanation does not explain how local changes affect the bulk dynamics.

In vitro imaging of the microtubule gives a way to observe it in real time and has been used extensively to measure and quantify the unique structural properties. The persistence lengths of our microtubules are within the 0.1–5.1 mm range

found in the literature showing that the microtubule is a very rigid structure with a persistence length that is order of magnitude higher than then its actual length [6]. Persistence length and Young's modulus give a quantifiable measurement of the rigidity of the microtubule and these are key measurements in understanding the bulk properties of the microtubule. However, they are not enough for extracting the underlying local interactions of the tubulin proteins. These local interactions, whether they are between taxol-bound tubulins, GTP-bound tubulins, GDP-bound tubulins, or a mixture of all three are the reason for not only the static, but also the dynamic mechanical properties of the microtubule.

The methodology presented takes the study of microtubules from a continuous model that assumes homogeneity to a meso-scale model of coupled oscillators that expand the methodology to lead to future improvements. It has been predicted theoretically that the MT has a phonon spectrum in the frequency regime of kHz to GHz. Most calculations fall in the GHz regime for longitudinal vibration modes and Atanasov predicted tranverse waves to be in the kHz regime [7–11]. The loss factor due to damping in the buffer solution has been estimated to be 0.01 up to 1000 [10]. Our experimental measurements show that the phonon spectrum of the bending modes of the microtubule lies in the lower predicted frequency regime.

The frequencies measured in the phonon spectrum and the speed at which we are able to image the microtubule affects the experimentally measured velocity. The predicted velocity of the wave along the helix direction of the microtubule is in the range of $147\text{--}167 \text{ m s}^{-1}$, and for longitudinal waves it is in the range of $1200\text{--}1300 \text{ m s}^{-1}$ [7]. The longitudinal wave velocities are higher due to the stronger bonding along a protofilament. The experimentally measured velocity here is due to the transverse waves and is smaller than the predicted velocities by other groups. The results presented here are $40\text{--}140 \text{ mm s}^{-1}$.

The microtubule has a fabulous structure. During its self assembly, every time a new dimer is attached to the lattice, there is a release of energy equal to $12 k_B T$ through GTP hydrolysis. A part of this energy is stored in the lattice as mechanical strain. Thus the microtubule behaves as a prestressed beam rather than a regular beam. A prestressed beam is more resistant to bending because when bent, the strained is released. When a regular beam is bent, it creates strain, that could lead to cracking. The phonon spectrum obtained here does not resemble the one of a regular beam for which the restoring force is proportional to the square of the displacement. It resembles more the one of a prestressed beam for which the force is linear with the displacement.

The limitations of the method proposed here assumes uniform and harmonic interaction between blocks, it measures only transversal modes, rather than all existing modes and it uses the thermal energy of the water to excite the modes, rather than providing a controlled excitation.

Future experiments must be done to excite and measure other modes of the microtubule. One has to go beyond thermal energy and engineer actuators to excite the longitudinal and torsional modes for a complete picture of local properties of the microtubule as well as to study the local differences between regular and cancerous microtubule or the effects and efficiency of different pharmaceutical compounds, such as chemotherapy drugs. On the other hand, the biological world, like the microtubule, can serve as inspiration for discovering new materials displaying radically new surface physics, such as dissipationless transport of mechanical energy and this provides the basis for a range of completely novel applications, such as bullet proof vests, submarine acoustic shielding, airplane radar shielding, and earthquake proofing. The field of topological mechanics is expanding to active, self-assembled materials in fluids and the microtubule is an example of such materials.

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